New Keynesian Model with Money

Assumptions

- Imperfect competition in the Market: Each firm produces a differentiated good for which it sets the Price.
- Only a fraction of firms can reset their prices in any given period.
- Assume a continuum of firms $i \in [0, 1]$. Each firm produces a differentiated good.

Households

Preferences and Budget Constraint

- The representative household's utility function is defined by:

$$U\left(C_t, N_t, \frac{M_t}{P_t}\right)$$

The marginal utilities reflect positive but diminishing returns to consumption and money holdings, and increasing marginal disutility from labor:

$$U_{c,t} = \frac{\partial U\left(C_t, N_t, \frac{M_t}{P_t}\right)}{\partial C_t} > 0;\ U_{cc,t} = \frac{\partial^2 U\left(C_t, N_t, \frac{M_t}{P_t}\right)}{\partial C_t^2} \leq 0$$

$$U_{n,t} = \frac{\partial U\left(C_t, N_t, \frac{M_t}{P_t}\right)}{\partial N_t} < 0;\ U_{nn,t} = \frac{\partial^2 U\left(C_t, N_t, \frac{M_t}{P_t}\right)}{\partial C_t^2} \leq 0$$

$$U_{m,t} = \frac{\partial U(C_t, N_t, m_t)}{\partial m_t} > 0;\ U_{mm,t} = \frac{\partial^2 U(C_t, N_t, m_t)}{\partial m_t^2} \leq 0$$

Furthermore, we assume additive separability in the utility function, meaning the cross-partial derivatives are zero:

$$U_{cn,t} = U_{cm,t} = U_{nm,t} = 0$$

- The aggregate consumption basket is defined by a Dixit-Stiglitz aggregator:

$$C_t = \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di\right)^{\frac{\varepsilon}{\varepsilon-1}}$$

  - ε is the elasticity of substitution between varieties.
- The household's nominal flow budget constraint is given by:

$$\int_0^1 P_t(i).C_t(i)\, di + Q_t B_t \leq B_{t-1} + W_t N_t + T_t$$

  - Q_t is the price of a one-period risk-free nominal bond B_t. W_t is the nominal wage. T_t represents lump-sum transfers.

Optimal Allocation of Consumptions Expenditures (Dual Problem)

- Given prices and the budget constraint, households must choose how to optimally allocate their consumption. Solving:

$$\max_{C_t(i)} C_t = \left[\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di\right]^{\frac{\varepsilon}{\varepsilon-1}}$$

$$s.t. \quad \int_0^1 P_t(i).C_t(i)\,di \leq Z_t$$

Setting up the Lagrangian:

$$L = \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}}\right)^{\frac{\varepsilon}{\varepsilon-1}} - \lambda\left(\int_0^1 P_t(i).C_t(i) - Z_t\right)$$

$$\frac{\partial L}{\partial C_{i,t}} \rightarrow \left(\frac{\varepsilon}{\varepsilon-1}\right)\left(\int_0^1 C_{i,t}{}^{\frac{\varepsilon-1}{\varepsilon}}\right)^{\frac{1}{\varepsilon-1}}\left(\frac{\varepsilon-1}{\varepsilon}\right)C_{i,t}^{-\frac{1}{\varepsilon}} - \lambda P_{i,t} = 0$$

$$\left[\underbrace{\left(\int_0^1 C_{i,t}{}^{\frac{\varepsilon-1}{\varepsilon}}\right)^{\frac{\varepsilon}{\varepsilon-1}}}_{C_t}\right]^{1/\varepsilon} C_{i,t}{}^{-\frac{1}{\varepsilon}} = \lambda P_{i,t}$$

$$C_t^{\frac{1}{\varepsilon}}.C_{i,t}{}^{-\frac{1}{\varepsilon}} = \lambda P_{i,t}, \quad \text{for all } i \in [0,1]$$

- Thus, for any two goods (i,j):

$$\frac{C_t^{\frac{1}{\varepsilon}}.C_{i,t}{}^{-\frac{1}{\varepsilon}}}{C_t^{\frac{1}{\varepsilon}}.C_{j,t}{}^{-\frac{1}{\varepsilon}}} = \frac{\lambda P_{i,t}}{\lambda P_{j,t}} \quad \rightarrow \quad C_{i,t}{}^{-\frac{1}{\varepsilon}} = \left(\frac{P_{i,t}}{P_{j,t}}\right) C_{j,t}{}^{-\frac{1}{\varepsilon}}$$

$$C_{i,t} = C_{j,t}\left(\frac{P_{i,t}}{P_{j,t}}\right)^{-\varepsilon} \qquad (1)$$

which can be substituted into the expression for consumption expenditures to yield:

$$Z_t = \int_0^1 P_t(i).C_t(i)\,di \quad \rightarrow \quad Z_t = \int_0^1 P_t(i).C_t(j)\left(\frac{P_t(i)}{P_t(j)}\right)^{-\varepsilon} di$$

$$Z_t = C_{j,t}.P_{j,t}^{\varepsilon}\int_0^1 P_{i,t}^{1-\varepsilon}\,di \quad \rightarrow \quad Z_t = C_{j,t}.P_{j,t}^{\varepsilon}\left[\left(\int_0^1 P_{i,t}{}^{1-\varepsilon}\,di\right)^{\frac{1}{1-\varepsilon}}\right]^{1-\varepsilon}$$

$$Z_t = C_{j,t}.P_{j,t}^{\varepsilon}.P_t^{1-\varepsilon} \quad \rightarrow \quad C_{j,t} = \frac{Z_t}{P_t}.\left(\frac{P_{j,t}}{P_t}\right)^{-\varepsilon} \quad \text{for all } i \in [0,1] \qquad (2)$$

- The latter condition can then be substituted into the definition of C_t to obtain:

$$C_t = \left(\int_0^1 C_{i,t}{}^{\frac{\varepsilon-1}{\varepsilon}}\,di\right)^{\frac{\varepsilon}{\varepsilon-1}} \quad \rightarrow \quad C_t = \left(\int_0^1 \left[\frac{Z_t}{P_t}.\left(\frac{P_t(i)}{P_t}\right)^{-\varepsilon}\right]^{\frac{\varepsilon-1}{\varepsilon}} di\right)^{\frac{\varepsilon}{\varepsilon-1}}$$

$$C_t = Z_t P_t^{\varepsilon-1}\left(\int_0^1 P_t(i)^{1-\varepsilon}dj\right)^{\frac{\varepsilon}{\varepsilon-1}} \quad \rightarrow \quad C_t = Z_t P_t^{\varepsilon-1}\left[\left(\int_0^1 P_t(i)^{1-\varepsilon}dj\right)^{\frac{1}{1-\varepsilon}}\right]^{-\varepsilon}$$

$$C_t = Z_t P_t^{\varepsilon-1} P_t^{-\varepsilon} \quad \rightarrow \quad Z_t = P_t C_t \qquad (3)$$

Combining the equation (2) and (3):

$$C_t(i) = \frac{Z_t}{P_t} \cdot \left(\frac{P_t(i)}{P_t}\right)^{-\varepsilon} \quad \rightarrow \quad C_t(i) = \frac{P_t C_t}{P_t}\left(\frac{P_t(i)}{P_t}\right)^{-\varepsilon}$$

$$C_t(i) = C_t\left(\frac{P_t(i)}{P_t}\right)^{-\varepsilon}$$

Note: The dual problem is important because by minimizing the cost of acquiring the consumption basket, we derive the downward-sloping demand curve for each individual good. This specific demand structure is what grants intermediate firms their monopolistic market power, allowing them to act as price-setters rather than price-takers.

Utility Maximization Problem

- The representative household maximize the objective function:

$$E_0\left[\sum_{t=0}^{\infty} \beta^t U\left(C_t, N_t, \frac{M_t}{P_t}\right)\right]$$

$$s.t. \quad \int_0^1 P_t(i)C_t(i)\,di + M_t + Q_t B_t \leq M_{t-1} + B_{t-1} + W_t N_t + T_t$$

- Generalizing, Households solve the following problem:

$$\max_{C_t, N_t, B_t, \frac{M_t}{P_t}} U = E_0\left[\sum_{t=0}^{\infty} \beta^t U(C_t, N_t, m_t)\right]$$

$$s.t \quad P_t C_t + M_t + Q_t B_t \leq M_{t-1} + B_{t-1} + W_t N_t + T_t$$

Setting up the Lagrangian:

$$L = E_0\left[\sum_{t=0}^{\infty} \beta^t U(C_t, N_t, m_t)\right] - \lambda_t(P_t C_t + M_t + Q_t B_t - M_{t-1} - B_{t-1} - W_t N_t - T_t)$$

$$\frac{\partial L}{\partial C_t} \rightarrow \beta^t U_{c,t} - \lambda_t P_t = 0 \quad (4)$$

$$\frac{\partial L}{\partial N_t} \rightarrow \beta^t U_{n,t} + \lambda_t W_t = 0 \quad (5)$$

$$\frac{\partial L}{\partial M_t} \rightarrow \beta^t U_{m,t}\frac{1}{P_t} - \lambda_t + E_t\{\lambda_{t+1}\} = 0 \quad (6)$$

$$\frac{\partial L}{\partial B_t} \rightarrow -\lambda_t Q_t + E_t\{\lambda_{t+1}\} = 0 \quad (7)$$

- From equation (7), we obtain:

$$\frac{E_t[\lambda_{t+1}]}{\lambda_t} = Q_t = \frac{1}{1 + i_t} \qquad (8)$$

- Using equation (4) and shifting one period forward, we obtain:

$$\beta^{t+1} U_{c,t+1} = E_t\{\lambda_{t+1} P_{t+1}\}$$

Then:

$$E_t\left[\frac{\beta^t U_{c,t}}{\beta^{t+1}U_{c,t+1}}\right] = E_t\left[\frac{\lambda_t P_t}{\lambda_{t+1}P_{t+1}}\right] \quad \rightarrow \quad E_t\left[\frac{U_{c,t}}{U_{c,t+1}}\right] = \beta E_t\left[\frac{\lambda_t P_t}{\lambda_{t+1}P_{t+1}}\right]$$

$$Q_t = \beta E_t\left[\frac{U_{c,t+1}}{U_{c,t}}\frac{P_t}{P_{t+1}}\right]$$

Note: This condition determines the optimal path of consumption and saving for the household. The household must be indifferent at the margin between consuming an additional unit of wealth today or using those funds to purchase a bond.

- Using equations (4) and (5), then:

$$\frac{\beta^t U_{c,t}}{\beta^t U_{n,t}} = \frac{\lambda_t P_t}{-\lambda_t W_t} \quad \rightarrow \quad -\frac{U_{n,t}}{U_{c,t}} = \frac{W_t}{P_t}$$

Note: This is the optimal intra-temporal condition (Intra-temporal Euler equation). The household equates its marginal rate of substitutions (MRS) between labor and consumption to the real wage. Intuitive, the marginal cost of working an additional hour is exactly compensated by the marginal benefit obtained from consuming the goods that can be purchased with the additional wage.

- From equations (4) and (6), we obtain:

$$\frac{\beta^t U_{c,t}}{\beta^t U_{m,t}} = E_t\left[\frac{\lambda_t P_t}{\lambda_t P_t - \lambda_{t+1}P_t}\right] \quad \rightarrow \quad \frac{U_{c,t}}{U_{m,t}} = E_t\left[\frac{1}{1-\frac{\lambda_{t+1}}{\lambda_t}}\right]$$

$$\frac{U_{c,t}}{U_{m,t}} = 1 - Q_t \quad \rightarrow \quad \frac{U_{m,t}}{U_{c,t}} = \frac{i_t}{1+i_t}$$

Note: This equation illustrates that the demand for real money balances depends positively on consumption (transaction motive) and negatively on the nominal interest rate. The term $\frac{i_t}{1+i_t}$ represents the opportunity cost of holding money instead of investing in interest-bearing bonds.

An Example with Separable Utility

- The household's utility function is assumed to have the functional form:

$$U\left(C_t, N_t, \frac{M_t}{P_t}\right) = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} + \frac{(M_t/P_t)^{1-v}}{1-v}$$

Replacing:

- Euler Equation:

$$1 = \beta E_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\sigma}(1+i_t)(1+\pi_t)\right]$$

- Labor Supply Equation:

$$C_t^{\sigma} N_t^{\varphi} = \frac{W_t}{P_t}$$

- Money Demand Equation:

$$\frac{M_t}{P_t} = C_t^{\sigma\backslash v}\left(\frac{1+i_t}{i_t}\right)^{-1/v}$$

Firms

- The function of production for each firm is represented by:

$$Y_t(i) = A_t N_t(i)^{1-\alpha}$$

  - A_t represented the level of technology.

Aggregate Price Level Dynamics

- Let $S(t) \subset [0,1]$ represent the set of firms not reoptimizing their posted price in period t and the fact that all firms resetting prices will choose an identical price P_t^*, then:

$$P_t = \left[\int_{s(t)}^{\cdot} P_t(i)^{1-\varepsilon}\, di + (1-\theta) P_t^{*1-\varepsilon}\right]^{\frac{1}{1-\varepsilon}}$$

$$P_t = \left[\theta P_{t-1}^{1-\varepsilon} + (1-\theta) P_t^{*1-\varepsilon}\right]^{\frac{1}{1-\varepsilon}}$$

  - P_t denotes the aggregate prices index. P_t^* is the price set in period t by firms reoptimizing their price in that period.

Dividing both sides by P_{t-1}:

$$P_t^{1-\varepsilon} = \theta P_{t-1}^{1-\varepsilon} + (1-\theta) P_t^{*1-\varepsilon}$$

$$\frac{P_t^{1-\varepsilon}}{P_{t-1}^{1-\varepsilon}} = \theta \frac{P_{t-1}^{1-\varepsilon}}{P_{t-1}^{1-\varepsilon}} + (1-\theta) \frac{P_t^{*1-\varepsilon}}{P_{t-1}^{1-\varepsilon}}$$

$$\Pi_t^{1-\varepsilon} = \theta + (1-\theta)\left(\frac{P_t^*}{P_{t-1}}\right)^{1-\varepsilon}$$

$$\boxed{\Pi_t = \frac{P_t}{P_{t-1}} = 1 + \pi_t}$$

- Notice that in a steady state with zero inflation $P_t^* = P_t = P_{t-1}$ for all t.

Optimal Price Setting

- It solves the problem:

$$\boxed{Q_{t,t+k} = \beta^k \left(\frac{C_{t+k}}{C_t}\right)^{-\sigma} \left(\frac{P_t}{P_{t+k}}\right)}$$

$$\max_{P_t^*} \sum_{k=0}^{\infty} \theta^k E_t \left[Q_{t,t+k}\left(P_t^* Y_{t+k\,|\,t} - \Psi_{t+k}(Y_{t+k\,|\,t})\right)\right]$$

$$s.t. \quad Y_{t+k\,|\,t} = \left(\frac{P_t^*}{P_{t+k}}\right)^{-\varepsilon} C_{t+k}, \quad \text{for } k = 0, 1, 2, \ldots$$

  - $Q_{t,t+k}$ is the stochastic discount factor for nominal payoffs. Ψ_t is the cost function. $Y_{t+k\,|\,t}$ denotes output in period $t+k$ for a firm that last reset its price in period t.

Solving:

$$[P_t^*] \rightarrow \sum_{k=0}^{\infty} \theta^k E_t \left[Q_{t,t+k}\left(Y_{t+k\,|\,t} + P_t^* \underbrace{\frac{\partial Y_{t+k\,|\,t}}{\partial P_t^*}}_{w} - \Psi'_{t+k}(Y_{t+k\,|\,t}) \underbrace{\frac{\partial Y_{t+k\,|\,t}}{\partial P_t^*}}_{w}\right)\right] = 0$$

Derivating w:

$$\frac{\partial Y_{t+k\,|\,t}}{\partial P_t^*} = -\varepsilon \frac{P_t^{*-\varepsilon-1}}{P_{t+k}^{-\varepsilon}} C_{t+k} = -\frac{\varepsilon}{P_t^*} \frac{P_t^{*-\varepsilon}}{P_{t+k}^{-\varepsilon}} C_{t+k} = -\varepsilon_t \frac{Y_{t+k\,|\,t}}{P_t^*} = 0$$

Replacing:

$$\sum_{k=0}^{\infty} \theta^k E_t \left[Q_{t,t+k} \left(Y_{t+k\,|\,t} + P_t^*(-\varepsilon_t) \frac{Y_{t+k\,|\,t}}{P_t^*} - \underbrace{\Psi'_{t+k}(Y_{t+k\,|\,t})}_{\psi_{t+k\,|\,t}} (-\varepsilon_t) \frac{Y_{t+k\,|\,t}}{P_t^*} \right) \right] = 0$$

$$\sum_{k=0}^{\infty} \theta^k E_t \left[Q_{t,t+k} Y_{t+k\,|\,t} \left(1 - \varepsilon \frac{P_t^* - \psi_{t+k\,|\,t}}{P_t^*} \right) \right] = 0$$

$$\sum_{k=0}^{\infty} \theta^k E_t \left[Q_{t,t+k} Y_{t+k\,|\,t} \left(\frac{P_t^* - \left(\frac{\varepsilon}{\varepsilon - 1}\right) \psi_{t+k\,|\,t}}{P_t^*} \right) \right] = 0$$

$$\sum_{k=0}^{\infty} \theta^k E_t \left[Q_{t,t+k} Y_{t+k\,|\,t} \left(P_t^* - \mathcal{M} \psi_{t+k\,|\,t} \right) \right] = 0$$

- $\psi_{t+k\,|\,t}$ is the nominal marginal cost in period $t+k$ for a firm which last reset its price in period t and $\mathcal{M} = \frac{\varepsilon}{\varepsilon - 1}$.

Note: Note that in the limiting case of no price rigidities ($\theta = 0$), the previous condition collapses to the familiar optimal price-setting condition under flexible prices:

$$P_t^* = \mathcal{M} \psi_{t\,|\,t}$$

which allows us to interpret M (frictionless markup) as the desired markup in the absence of constraints on the frequency of price adjustment.

- If we divide by P_{t-1} and letter $\Pi_{t,t+k} \equiv \frac{P_{t+k}}{P_t}$, we can rewrite as:

$$\sum_{k=0}^{\infty} \theta^k E_t \left[Q_{t,t+k} Y_{t+k\,|\,t} \left(\frac{P_t^*}{P_{t-1}} - \mathcal{M} MC_{t+k\,|\,t} \Pi_{t-1,t+k} \right) \right] = 0$$

- $MC_{t+k\,|\,t} = \psi_{t\,|\,t} / P_{t+k}$ is the real marginal cost in period $t+k$ for a firm whose price was last set in period t.

Equilibrium

- Market Clearing in the goods market:

$$Y_t(i) = C_t(i), \quad \text{for all } i \in [0,1]$$

Integrating across all goods:

$$Y_t = C_t$$

Substituting the household's optimal demand for good i into the market clearing condition:

$$Y_t(i) = Y_t \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon}$$

- To find aggregate employment, we integrate labor demand across all firms. Using the firm's production function $Y_t(i) = A_t N_t(i)^{1-\alpha}$, the labor demand for firm i is:

$$N_t(i) = \left(\frac{Y_t(i)}{A_t} \right)^{\frac{1}{1-\alpha}}$$

Integrating over all $i \in [0,1]$ yields aggregate employment N_t:

$$N_t = \int_0^1 N_t(i)di = \int_0^1 \left(\frac{Y_t(i)}{A_t}\right)^{\frac{1}{1-\alpha}} di$$

Substituting the demand function $Y_t(i)$ into the aggregate employment equation:

$$N_t = \int_0^1 \left(\frac{Y_t\left(\frac{P_t(i)}{P_t}\right)^{-\varepsilon}}{A_t}\right)^{\frac{1}{1-\alpha}} di = \left(\frac{Y_t}{A_t}\right)^{\frac{1}{1-\alpha}} \int_0^1 \left(\frac{P_t(i)}{P_t}\right)^{-\frac{\varepsilon}{1-\alpha}} di$$

Let V_t define the index of price dispersion in the economy:

$$V_t = \int_0^1 \left(\frac{P_t(i)}{P_t}\right)^{-\frac{\varepsilon}{1-\alpha}} di$$

We can then express the aggregate production relationship as:

$$N_t = \left(\frac{Y_t}{A_t}\right)^{\frac{1}{1-\alpha}} V_t \quad \rightarrow \quad Y_t = \frac{A_t N_t^{1-\alpha}}{V_t^{1-\alpha}}$$

Note: In a neighborhood of a zero-inflation steady state, the price dispersion index V_t is equal to 1 up to a first-order approximation.

- Aggregate Real Marginal Cost:

$$mc_t = \frac{W_t/P_t}{(1-\alpha)A_t N_t^{-\alpha}} = \frac{W_t/P_t}{MPN_t}$$

- Bonds Market Clearing:

$$B_t = 0 \quad \text{for all } t$$

- Finally, substituting the aggregate clearing condition ($C_t = Y_t$) into the household's Euler equation yields the non-linear Dynamic IS curve:

$$C_t^{-\sigma} = \beta E_t[C_{t+1}^{-\sigma}(1+i_t)(1+\pi_{t+1})^{-1}]$$

- The aggregate real wage is equated to the aggregate marginal rate of substitution:

$$\frac{W_t}{P_t} = N_t^{\varphi} C_t^{\sigma}$$

- And the money demand:

$$\frac{M_t}{P_t} = C_t^{\sigma/v}\left(\frac{1+i_t}{i_t}\right)^{-1/v}$$

Steady State

We consider a zero-inflation steady state, which implies that the aggregate price level is constant ($\Pi = 1 \rightarrow \pi = 0$). We also assume that there are no technological shocks ($A = 1$) and all variables are constant over time.

- Thus:
  - From the non-linear Dynamic IS curve (Euler equation):

$$C^{-\sigma} = \beta E_t[C^{-\sigma}(1+i)(1+\pi)^{-1}]$$

Since consumption is constant in steady state, the marginal utilities cancel out:

$$1 = \beta(1+i)(1+0)^{-1}$$

$$1 + i = \frac{1}{\beta}$$

Given zero inflation, the nominal interest rate equals the real interest rate, uniquely determined by the discount factor:

$$i = r = \frac{1-\beta}{\beta} \equiv \rho$$

  - Evaluating the non-linear optimal price condition under these steady-state values yields:

$$MC = \frac{1}{\mathcal{M}} = \frac{\varepsilon - 1}{\varepsilon}$$

  - Using the market clearing condition ($Y = C$) and the steady-state aggregate production relationship:

$$Y = N^{1-\alpha} \rightarrow N = Y^{\frac{1}{1-\alpha}}$$

Substituting this into the steady-state labor supply conditian:

$$\frac{W}{P} = \left(Y^{\frac{1}{1-\alpha}}\right)^{\varphi} Y^{\sigma} = Y^{\frac{\varphi}{1-\alpha}} Y^{\sigma} = Y^{\frac{\varphi+\sigma(1-\alpha)}{1-\alpha}}$$

Now, substituting the steady-state real wage and employment into the Aggregate Real Marginal Cost equation:

$$MC = \frac{Y^{\frac{\varphi+\sigma(1-\alpha)}{1-\alpha}}}{(1-\alpha)(1)\left(Y^{\frac{1}{1-\alpha}}\right)^{-\alpha}} = \frac{Y^{\frac{\varphi+\sigma(1-\alpha)}{1-\alpha}}}{(1-\alpha)Y^{\frac{-\alpha}{1-\alpha}}} = \frac{Y^{\frac{\varphi+\alpha+\sigma(1-\alpha)}{1-\alpha}}}{1-\alpha}$$

Equating this to the steady-state marginal cost from the price-setting condition $\left(MC = \frac{1}{\mathcal{M}}\right)$ and solving for Output:

$$\frac{Y^{\frac{\varphi+\alpha+\sigma(1-\alpha)}{1-\alpha}}}{1-\alpha} = \frac{1}{\mathcal{M}}$$

$$Y = \left(\frac{1-\alpha}{\mathcal{M}}\right)^{\frac{1-\alpha}{\varphi+\alpha+\sigma(1-\alpha)}}$$

Log-linearization of the model

Aggregate Real Wage

$$\frac{W_t}{P_t} = N_t^{\varphi} C_t^{\sigma}$$

$$\ln(W_t) - \ln(P_t) = \varphi \ln(N_t) + \sigma \ln(C_t)$$

$$\frac{dW_t}{W} - \frac{dP_t}{P} = \varphi \frac{dN_t}{N} + \sigma \frac{dC_t}{C}$$

$$\tilde{w}_t - \tilde{p}_t = \varphi \tilde{n}_t + \sigma \tilde{c}_t$$

Euler Equation

$$C_t^{-\sigma} = \beta E_t[(1+i_t)(1+p)^{-1}(1+\pi_{t+1})^{-1} C_{t+1}^{-\sigma}]$$

$$-\sigma \ln(C_t) = \ln(\beta) + \ln(1+i_t) - \ln(1+p) - E_t \ln(1+\pi_{t+1}) - \sigma E_t \ln(C_{t+1})$$

$$-\sigma \frac{dC_t}{C} = \frac{di_t}{(1+i)} - \frac{d\pi_{t+1}}{(1+\pi)} - \sigma \frac{dC_{t+1}}{C}$$

$$\sigma E_t \tilde{c}_{t+1} - \sigma \tilde{c}_t = \tilde{\imath}_t - E_t \tilde{\pi}_{t+1}$$

$$\tilde{c}_t = E_t \tilde{c}_{t+1} - \frac{1}{\sigma}(\tilde{\imath}_t - E_t \tilde{\pi}_{t+1})$$

Money Demand

$$\frac{M_t}{P_t} = C_t^{\sigma \backslash v} \left(\frac{1+i_t}{i_t}\right)^{-1/v}$$

- By taking logs, applying a first-order Taylor approximation around the steady-state interest rate i_t, and assuming a unit elasticity of money demand with respect to output:

$$\tilde{m}_t - \tilde{p}_t = \tilde{c}_t - \eta \tilde{\imath}_t$$

$$\tilde{m}_t - \tilde{p}_t = \tilde{y}_t - \eta \tilde{\imath}_t$$

Goods Market Clearing

$$Y_t = C_t$$

$$\ln(Y_t) = \ln(C_t)$$

$$\frac{dY_t}{Y} = \frac{dC_t}{C}$$

$$\tilde{y}_t = \tilde{c}_t$$

Aggregate Product Relationship

$$Y_t = \frac{A_t N_t^{1-\alpha}}{V_t^{1-\alpha}}$$

$$\ln(Y_t) = \ln(A_t) + (1-\alpha)\ln(N_t) - (1-\alpha)\ln(V_t)$$

$$\frac{dY_t}{Y} = \frac{dA_t}{A} + (1-\alpha)\frac{dN_t}{N} - (1-\alpha)\frac{dV_t}{V}$$

$$\tilde{y}_t = \tilde{a}_t + (1-\alpha)\tilde{n}_t - (1-\alpha)\tilde{v}_t$$

$\tilde{v}_t \approx 0$, in a neighborhood of the zero-inflation steady state:

$$\tilde{y}_t = \tilde{a}_t + (1-\alpha)\tilde{n}_t$$

Aggregate Real Marginal Cost

$$MC_t = \frac{W_t/P_t}{(1-\alpha)A_t N_t^{-\alpha}} = \frac{W_t/P_t}{MPN_t}$$

$$\ln MC_t = \ln W_t - \ln P_t - \ln(1-\alpha) - \ln A_t + \alpha \ln N_t$$

$$\widetilde{mc}_t = (\tilde{w}_t - \tilde{p}_t) - (\tilde{a}_t - \alpha \tilde{n}_t)$$

Replacing by the aggregate product relationship:

$$\widetilde{mc}_t = (\tilde{w}_t - \tilde{p}_t) - (\tilde{a}_t - \alpha \tilde{n}_t)$$

$$\widetilde{mc}_t = (\tilde{w}_t - \tilde{p}_t) - \left(\tilde{a}_t - \alpha\left(\frac{\tilde{y}_t - \tilde{a}_t}{1-\alpha}\right)\right)$$

$$\widetilde{mc}_t = (\tilde{w}_t - \tilde{p}_t) - \frac{1}{1-\alpha}(\tilde{a}_t - \alpha \tilde{y}_t)$$

In a way consistent, we can:

$$\widetilde{mc}_{t+k\,|\,t} = (\tilde{w}_{t+k} - \tilde{p}_{t+k}) - \frac{1}{1-\alpha}\left(\tilde{a}_{t+k} - \alpha \tilde{y}_{t+k\,|\,t}\right)$$

$$\widetilde{mc}_{t+k\,|\,t} - \widetilde{mc}_{t+k} = (\tilde{w}_{t+k} - \tilde{p}_{t+k}) - \frac{1}{1-\alpha}\left(\tilde{a}_{t+k} - \alpha \tilde{y}_{t+k\,|\,t}\right) - \widetilde{mc}_{t+k}$$

$$\widetilde{mc}_{t+k\,|\,t} = \widetilde{mc}_{t+k} + \frac{\alpha}{1-\alpha}\left(\tilde{y}_{t+k\,|\,t} - \tilde{y}_{t+k}\right)$$

$$\widetilde{mc}_{t+k\,|\,t} = \widetilde{mc}_{t+k} - \frac{\alpha\varepsilon}{1-\alpha}(p_t^* - p_{t+k})$$

Aggregate Price Level

$$P_t = \left[\theta P_{t-1}^{1-\varepsilon} + (1-\theta){P_t^*}^{1-\varepsilon}\right]^{\frac{1}{1-\varepsilon}}$$

Taking logs and totally differentiating around the zero-inflation steady state:

$$(1-\varepsilon)\frac{dP_t}{P} = \theta(1-\varepsilon)\frac{dP_{t-1}}{P} + (1-\theta)(1-\varepsilon)\frac{dP_t^*}{P}$$

$$\tilde{p}_t = \theta \tilde{p}_{t-1} + \tilde{p}_t^* - \theta \tilde{p}_t^*$$

$$\tilde{p}_t - \tilde{p}_{t-1} = \theta \tilde{p}_{t-1} + \tilde{p}_t^* - \theta \tilde{p}_t^* - \tilde{p}_{t-1}$$

$$\tilde{\pi}_t = (1-\theta)(\tilde{p}_t^* - \tilde{p}_{t-1})$$

Optimal Price Setting

$$\sum_{k=0}^{\infty} \theta^k E_t\left[Q_{t,t+k} Y_{t+k\,|\,t}\left(\frac{P_t^*}{P_{t-1}} - \mathcal{M}MC_{t+k\,|\,t}\Pi_{t-1,t+k}\right)\right] = 0$$

Using Taylor Approximation Rule:

$$Y\sum_{k=0}^{\infty} \theta^k \beta^k E_t\left[(1 + \tilde{p}_t^* - \tilde{p}_{t-1}) - \left(1 + \widetilde{mc}_{t+k\,|\,t}\right)(1 + \tilde{p}_{t+k} - \tilde{p}_{t-1})\right] = 0$$

$$\sum_{k=0}^{\infty} (\theta\beta)^k E_t\left[\tilde{p}_t^* - \tilde{p}_{t-1} - \widetilde{mc}_{t+k\,|\,t} - (\tilde{p}_{t+k} - \tilde{p}_{t-1})\right] = 0$$

The term $\tilde{p}_t^* - \tilde{p}_{t-1}$ not depends of k, therefore we can:

$$\sum_{k=0}^{\infty} \theta^k \beta^k (\tilde{p}_t^* - \tilde{p}_{t-1}) - \sum_{k=0}^{\infty} (\theta\beta)^k E_t\left[\widetilde{mc}_{t+k\,|\,t} + (\tilde{p}_{t+k} - \tilde{p}_{t-1})\right] = 0$$

$$(\tilde{p}_t^* - \tilde{p}_{t-1}) \sum_{k=0}^{\infty} (\theta\beta)^k = \sum_{k=0}^{\infty} (\theta\beta)^k E_t\left[\widetilde{mc}_{t+k\,|\,t} + (\tilde{p}_{t+k} - \tilde{p}_{t-1})\right]$$

Using infinite geometric series propierties:

$$(\tilde{p}_t^* - \tilde{p}_{t-1}) \left(\frac{1}{1-\theta\beta}\right) = \sum_{k=0}^{\infty} (\theta\beta)^k E_t\left[\widetilde{mc}_{t+k\,|\,t} + (\tilde{p}_{t+k} - \tilde{p}_{t-1})\right]$$

$$(\tilde{p}_t^* - \tilde{p}_{t-1}) = (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k E_t\left[\widetilde{mc}_{t+k\,|\,t} + (\tilde{p}_{t+k} - \tilde{p}_{t-1})\right]$$

Replacing the individual marginal cost with the aggregate marginal cost relationship derived previously:

$$(\tilde{p}_t^* - \tilde{p}_{t-1}) = (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k E_t\left[\widetilde{mc}_{t+k} - \frac{\alpha\varepsilon}{1-\alpha}(\tilde{p}_t^* - \tilde{p}_{t+k}) + (\tilde{p}_{t+k} - \tilde{p}_{t-1})\right]$$

Adding and subtracting $\frac{\alpha\varepsilon}{1-\alpha}\tilde{p}_{t-1}$ inside the expectation to group terms:

$$(\tilde{p}_t^* - \tilde{p}_{t-1}) = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k E_t\left[\widetilde{mc}_{t+k} - \frac{\alpha\varepsilon}{1-\alpha}(\tilde{p}_t^* - \tilde{p}_{t-1}) - \frac{\alpha\varepsilon}{1-\alpha}\tilde{p}_{t-1} + \frac{\alpha\varepsilon}{1-\alpha}\tilde{p}_{t-1} + (\tilde{p}_{t+k} - \tilde{p}_{t-1})\right]$$

$$(\tilde{p}_t^* - \tilde{p}_{t-1}) = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k E_t\left[\widetilde{mc}_{t+k} - \frac{\alpha\varepsilon}{1-\alpha}(\tilde{p}_t^* - \tilde{p}_{t-1}) + \left(1 + \frac{\alpha\varepsilon}{1-\alpha}\right)(\tilde{p}_{t+k} - \tilde{p}_{t-1})\right]$$

Let $\frac{1}{\Theta} = \frac{1-\alpha+\alpha\varepsilon}{1-\alpha}$, and pulling the constant term out of the sum:

$$\tilde{p}_t^* - \tilde{p}_{t-1} + \frac{\alpha\varepsilon}{1-\alpha}(\tilde{p}_t^* - \tilde{p}_{t-1}) = (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k E_t\left[\widetilde{mc}_{t+k} + \frac{1}{\Theta}(\tilde{p}_{t+k} - \tilde{p}_{t-1})\right]$$

$$\left(1 + \frac{\alpha\varepsilon}{1-\alpha}\right)(\tilde{p}_t^* - \tilde{p}_{t-1}) = (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k E_t\left[\widetilde{mc}_{t+k} + \frac{1}{\Theta}(\tilde{p}_{t+k} - \tilde{p}_{t-1})\right]$$

Replacing:

$$\frac{1}{\Theta}(\tilde{p}_t^* - \tilde{p}_{t-1}) = (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k E_t\left[\widetilde{mc}_{t+k} + \frac{1}{\Theta}(\tilde{p}_{t+k} - \tilde{p}_{t-1})\right]$$

$$\tilde{p}_t^* - \tilde{p}_{t-1} = (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k E_t[\Theta\widetilde{mc}_{t+k} + (\tilde{p}_{t+k} - \tilde{p}_{t-1})]$$

Splitting the summation into two parts:

$$\tilde{p}_t^* - \tilde{p}_{t-1} = (1-\theta\beta)\sum_{k=0}^{\infty}(\theta\beta)^k E_t[\Theta\widetilde{mc}_{t+k}] + (1-\theta\beta)\sum_{k=0}^{\infty}(\theta\beta)^k E_t[\tilde{p}_{t+k} - \tilde{p}_{t-1}]$$

Using the property of the sum of differences, the second term collapses to expected inflation:

$$\tilde{p}_t^* - \tilde{p}_{t-1} = (1-\beta\theta)\Theta\sum_{k=0}^{\infty}(\beta\theta)^k E_t[\widetilde{mc}_{t+k}] + \sum_{k=0}^{\infty}(\beta\theta)^k E_t[\tilde{\pi}_{t+k}]$$

To express this recursively (quasi-differencing):

$$\tilde{p}_t^* - \tilde{p}_{t-1} = \beta\theta E_t[\tilde{p}_{t+1}^* - \tilde{p}_t] + (1-\beta\theta)\Theta\widetilde{mc}_t + \tilde{\pi}_t$$

New Keynesian Phillips Curve (NKPC)

- Combining the recursive optimal price equation with the log-linearized Aggregate Price Level:

$$\frac{1}{1-\theta}\tilde{\pi}_t = \beta\theta E_t\left[\frac{1}{1-\theta}\tilde{\pi}_{t+1}\right] + (1-\beta\theta)\Theta\widetilde{mc}_t + \tilde{\pi}_t$$

$$\frac{\theta}{1-\theta}\tilde{\pi}_t = \frac{\beta\theta}{1-\theta}E_t[\tilde{\pi}_{t+1}] + (1-\beta\theta)\Theta\widetilde{mc}_t$$

$$\tilde{\pi}_t = \beta E_t[\tilde{\pi}_{t+1}] + \frac{(1-\beta\theta)(1-\theta)}{\theta}\Theta\widetilde{mc}_t$$

Defining $\lambda \equiv \frac{(1-\beta\theta)(1-\theta)}{\theta}\Theta$, we obtain:

$$\tilde{\pi}_t = \beta E_t[\tilde{\pi}_{t+1}] + \lambda\widetilde{mc}_t$$

Average real marginal cost can be expressed as:

$$MC_t = \frac{W_t/P_t}{MPN_t}$$

$$\widetilde{mc}_t = (\tilde{w}_t - \tilde{p}_t) - \widetilde{mpn}_t$$

Replacing the real wage from the labor supply ($\tilde{w}_t - \tilde{p}_t = \sigma\tilde{y}_t + \varphi\tilde{n}_t$) and the marginal product of labor ($\widetilde{mc}_t = \tilde{a}_t - \alpha\tilde{n}_t$):

$$\widetilde{mc}_t = (\sigma\tilde{y}_t + \varphi\tilde{n}_t) - (\tilde{a}_t - \alpha\tilde{n}_t)$$

$$\widetilde{mc}_t = \sigma\tilde{y}_t + (\varphi+\alpha)\tilde{n}_t - \tilde{a}_t$$

Substituting the labor demand derived from the aggregate production function $\left(\tilde{n}_t = \frac{1}{1-\alpha}(\tilde{y}_t - \tilde{a}_t)\right)$:

$$\widetilde{mc}_t = \sigma\tilde{y}_t + \left(\frac{\varphi+\alpha}{1-\alpha}\right)(\tilde{y}_t - \tilde{a}_t) - \tilde{a}_t$$

$$\widetilde{mc}_t = \left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right)\tilde{y}_t - \left(1 + \frac{\varphi+\alpha}{1-\alpha}\right)\tilde{a}_t$$

$$\widetilde{mc}_t = \left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right)\tilde{y}_t - \left(\frac{1+\varphi}{1-\alpha}\right)\tilde{a}_t$$

Flexible Price Equilibrium and the Natural Level of Output

- Under flexible prices, the real marginal cost is constant at its desired frictionless level, meaning its log-deviation from the steady state is zero ($\widetilde{mc}_t = 0$). And defining the natural level of output ($\tilde{y}_t^n$) as the equilibrium level of output under perfectly flexible prices. Evaluating equation under flexible prices:

$$0 = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha}\right)\tilde{y}_t^n - \left(\frac{1 + \varphi}{1 - \alpha}\right)\tilde{a}_t$$

$$\tilde{y}_t^n = \left(\frac{1 + \varphi}{(1 - \alpha)\sigma + \varphi + \alpha}\right)\tilde{a}_t = \psi_{ya}^n \tilde{a}_t$$

The Output Gap and the NKPC

- Define the Output Gap ($\tilde{x}_t$) as the log deviation of output from its natural level: $\tilde{x}_t = \tilde{y}_t - \tilde{y}_t^n$. Subtracting equations, yields the marginal cost in terms of the output gap:

$$\widetilde{mc}_t - 0 = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha}\right)(\tilde{y}_t - \tilde{y}_t^n)$$

$$\widetilde{mc}_t = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha}\right)(\tilde{x}_t)$$

Substitute this relationship back into the previously derived New Keynesian Phillips Curve:

$$\tilde{\pi}_t = \beta E_t[\tilde{\pi}_{t+1}] + \lambda \widetilde{mc}_t$$

$$\tilde{\pi}_t = \beta E_t[\tilde{\pi}_{t+1}] + \lambda\left(\sigma + \frac{\varphi + \alpha}{1 - \alpha}\right)(\tilde{x}_t)$$

Defining $\kappa \equiv \lambda\left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right)$, we obtain the final NKPC in terms of the output gap:

$$\tilde{\pi}_t = \beta E_t[\tilde{\pi}_{t+1}] + \kappa \tilde{x}_t$$

Dynamic IS Curve (DIS) in terms of Output Gap

- To maintain consistency, we must also express the household's Euler Equation in terms of the output gap:

$$\tilde{y}_t = E_t \tilde{y}_{t+1} - \frac{1}{\sigma}(\tilde{\imath}_t - E_t \tilde{\pi}_{t+1})$$

$$\tilde{y}_t - \tilde{y}_t^n = E_t[\tilde{y}_{t+1} - \tilde{y}_{t+1}^n] - \frac{1}{\sigma}(\tilde{\imath}_t - E_t \tilde{\pi}_{t+1}) + (E_t \tilde{y}_{t+1} - \tilde{y}_t^n)$$

$$\tilde{x}_t = E_t[\tilde{x}_{t+1}] - \frac{1}{\sigma}[\tilde{\imath}_t - E_t \tilde{\pi}_{t+1} - \sigma(E_t \tilde{y}_{t+1} - \tilde{y}_t^n)]$$

Define the natural rate of interest ($\tilde{r}_t^n$) as the real interest rate that prevails in the flexible price equilibrium:

$$\tilde{r}_t^n \equiv \sigma(E_t \tilde{y}_{t+1} - \tilde{y}_t^n) = \sigma \psi_{ya}^n E_t[\tilde{a}_{t+1} - \tilde{a}_t]$$

Substituting the natural rate of interest, we get the final ***Dynamic IS (DIS) equation***:

$$\tilde{x}_t = E_t[\tilde{x}_{t+1}] - \frac{1}{\sigma}[\tilde{\imath}_t - E_t \tilde{\pi}_{t+1} - \tilde{r}_t^n]$$